Status	Finished
Started	Wednesday, 21 October 2026, 4:44 PM
Completed	Wednesday, 21 October 2026, 4:54 PM
Duration	10 mins 14 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Binary Search
- c. Merge Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Removing duplicates from a list
- b. Arranging data in a specific order
- c. Merging two datasets
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Binary Search
c. Bubble Sort
d. Linear Search

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By comparing elements sequentially
c. By sorting data in a single pass
d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Quick Sort
b. Bubble Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Sorting
c. Rearranging
d. Searching

The correct answer is: Sorting

Question **7**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Sorting data in a non-sequential manner
b. Being the most efficient sorting algorithm
c. Bubbling up the largest element to its correct position with each pass
d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **8**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Divide and conquer
b. Dynamic programming
c. Backtracking
d. Greedy method

The correct answer is:
Divide and conquer

Question **9**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It increases the number of comparisons needed
b. It allows for faster searching
c. It makes the search operation slower
d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **10**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching
b. Sorting
c. Merging
d. Complexity

The correct answer is:
Complexity

Question **11**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Merge Sort
c. Bubble Sort
d. Binary Search

The correct answer is: Merge Sort

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are combined to form a single sorted list
c. They are discarded
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **13**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It makes it easier to select items based on their relationship to the rest of the items
c. It slows down the process
d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Bubble Sort
c. Linear Search
d. Quick Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It bubbles up the largest element in each pass
b. It recursively sorts subproblems
c. It divides the list into sublists
d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass